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The stellar interferometer at Narrabri, Australia

I. The principles of the intensity interferometer; general description

II. Measurement of the correlation

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In 1956 Hanbury Brown and Twiss demonstrated that the fluctuations in the outputs of two photomultipliers, illuminated by coherent light beams, are correlated. This noted experiment was an important step in applying the intensity interferometer — which had been developed in radio astronomy a few years earlier — to visible stars.

Since that time Hanbury Brown, Twiss and their co-workers have developed and built the optical intensity interferometer at Narrabri, with which it is possible to measure the angular sizes of a number of stars for the first time. This is done with the aid of a refined electronic apparatus, the "correlator", which measures the correlation between the fluctuations in the outputs of two photomultipliers. This electronic correlator has been designed and built by Mullard Research Laboratories.

The present subject is presented in two articles. The first gives an introduction to intensity interferometry and a general description of the apparatus in Narrabri, the second describes the correlator in some detail.

We are pleased to welcome Prof. Hanbury Brown, who wrote the first article, as a guest author to our Review.

I. The principles of the intensity interferometer; general description

R. Hanbury Brown

Introduction

The purpose of the stellar interferometer at Narrabri is to measure the angular diameter of the bright stars. The "angular diameter" of a star is the angle subtended by the diameter of the star at the Earth.

At first sight it seems that the simplest way to measure the angular diameter of a star would be to view it with a telescope and to measure the image. In fact that is precisely how the angular diameters of the Sun, the Moon and the planets are measured. However, in practice this method cannot be used for the stars because their distances are so great that their outlines cannot be seen in a telescope. There are two principal reasons why this is so. Firstly, the ability of a telescope to distinguish between a point source of light and a source of finite angular size is limited fundamentally by the size of the principal lens or mirror of the telescope and by the wavelength of light. Thus if we look at a point source of light with a telescope the image obtained will appear, not as a point, but as a bright disc surrounded by bright rings. If the principal mirror has a diameter of d metres the angular diameter of the disc, i.e. the *apparent* angular size of the point source, is approximately λ/d radians, where λ is the wavelength of the light. Thus, in theory we cannot with a telescope of diameter d measure the angular diameter Θ of a star unless Θ is greater than λ/d radians.

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The largest angular diameter of any star is that of Betelgeuse, 0.047 seconds of arc, and just to distinguish this star from a point source we should theoretically need a telescope lens or mirror roughly 3 metres in diameter. The *hot* bright stars have angular sizes considerably less than that of Betelgeuse; typically they subtend angles of 0.001 seconds of arc and we should require telescope mirrors with a diameter of about 150 metres to resolve them.

However, in making this estimate we have neglected a second limitation which is set by scintillation in the Earth's atmosphere. Irregularities in the atmosphere introduce random changes in the phase and amplitude of the light reaching the ground and, roughly speaking, these changes are uncorrelated at two points separated by more than about 10 cm. In practice this means that different parts of a large telescope mirror view the star through different irregularities and in consequence the image of a star is distorted, dances about, and is increased in size. The limitations set by the atmosphere depend on a variety of factors such as the height of the Observatory and the weather; roughly speaking they set limits to the minimum angular size which can be seen clearly in a telescope of between 1 and 0.1 seconds of arc. Thus, no matter how large our telescope, we are prevented by the Earth's atmosphere from measuring the angular size of stars.

There are, however, two types of instrument with which the angular diameters of stars have been measured. In both the light is received, not on one huge mirror with a diameter of λ/Θ, but on two comparatively small mirrors which can be separated by a distance λ/Θ. The first of these types is the well-known *Michelson interferometer*. Here the light received at the two small mirrors is brought together at the centre to give interference patterns. The second type is the *intensity interferometer* which has been developed in the past ten years. In this instrument the light received at two separated mirrors is focused on to two photoelectric detectors and, in effect, is converted into electric currents. The two currents are then brought together and "interfere" electrically; in other words the currents contain high-frequency fluctuations which are compared or correlated.

The stellar interferometer at Narrabri, the subject of the present two articles, is an intensity interferometer. In the first article we shall review the general principles on which it works and compare it with Michelson's interferometer. In the second article the *electronic correlator* which compares the currents from the two photoelectric detectors will be described. The correlator, which was designed and built at the Mullard Research Laboratories, presented considerable design problems.

The purpose of the measurements

The majority of the material in the Universe is condensed into stars and an understanding of their structure and evolution is one of the major aims of astronomy. Theoretical studies show that the mass and chemical composition of a star determine to a large extent its other properties and it is the aim of these studies to relate these basic parameters to the observed properties of the star. Two important properties of a star which can be observed are its *surface temperature* and its *size*, both of which exhibit a considerable variety. The fact that stars differ in temperature can be seen clearly from their spectra which, for historical reasons, are classified in the curious order *O*, *B*, *A*, *F*, *G*, *K*, *M* in descending order of temperature. The associated scale of temperature ranges from about 30 000 degrees for stars of type *O*, to 3000 degrees for type *M*. The physical size of stars also varies greatly. If we confine our attention to the most common types of star, known to astronomers as main sequence stars, their physical diameters vary from roughly 20 times that of the Sun (type *G*) for stars of type *O*, to about one half the size of the Sun for stars of type *M*. However, much greater variations in size are found if we look at rarer types of star. For example, "supergiant stars" of type *M* are about 500 times as large as the Sun, while at the other extreme the size of the "white dwarfs" is believed to be about 100 times smaller than the Sun.

Fig. 1 shows how the observed properties of stars can be displayed in a significant form. This diagram, called a Hertzsprung-Russell diagram, shows the relationship between surface temperature, spectral type, physical radius and total radiated light flux for different types of star. The spectral type and the associated surface temperature are shown along the horizontal axis. The vertical scale is a measure of the total light radiated by the star. The total light is measured in terms of the "absolute magnitude" of the star which is the apparent magnitude of the star if placed at a standard distance of 10 parsecs (32.6 light years) from the earth. Of course, the absolute magnitude cannot be measured directly but can be calculated from observations of the distance and the apparent magnitude [1]. The straight lines across the figure represent the physical radii of the stars relative to the Sun.

Now the data in fig. 1 are based on measurements of light flux, spectra, and distance (from the parallax). The temperature corresponding to each spectral type can be estimated from the general distribution of light flux with wavelength, or in other words from the colour; alternatively it can be found from a detailed analysis of the spectral lines and hence the degree of ionization of various elements in the stellar atmosphere. Generally

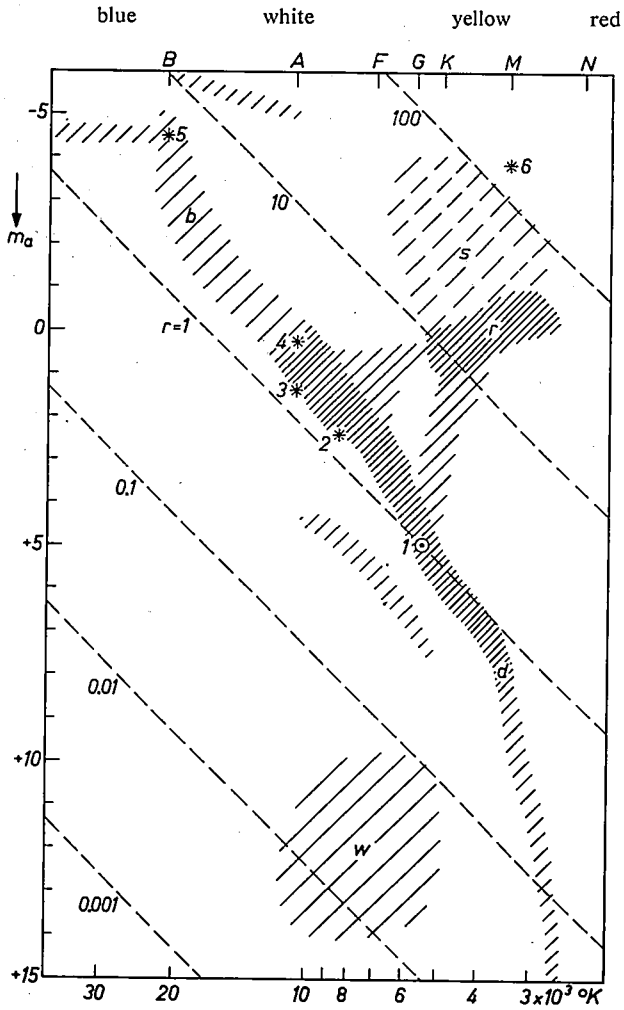


Fig. 1. The Hertzsprung-Russell-diagram, a classification of the stars according to absolute magnitude m_a (proportional to the log of the total radiated light flux) and surface temperature T. Lines of constant radius r are drawn according to Stefan's law (total radiated light flux $\propto T^4r^2$), valid for black body radiators. The values of r are relative to the radius of the sun. In the horizontal direction the stars are plotted according to their spectral type O, B, A, . . ., which is closely related to T. This relation is rather uncertain for hot stars.

Most stars lie on the "main sequence" (*b-d*), which includes the blue giants (*b*) and the red dwarfs (*d*). Large groups outside the main sequence are: the red giants (*r*), the super giants (*s*), and the white dwarfs (*w*). Some well-known stars are indicated: *1* Sun, *2* Altair, *3* Sirius, *4* Vega, *5* β Crucis, *6* Betelgeuse.

speaking both these methods are approximate but work reasonably well for stars cooler than type *A*. For hotter stars of type *O* and *B* there are two principal difficulties. Firstly, these hot stars are extremely blue and so the majority of their radiation is in the ultra-violet and is strongly absorbed by the earth's atmosphere; hence it is impossible, without observation from above the atmosphere, to find the true distribution of flux with wavelength. Secondly, their temperatures are so high that they show only a few weak spectral lines. As a result the temperatures of these very hot stars are uncertain.

The physical size of a star can be estimated from the flux of light received (apparent magnitude), the measured distance and the estimated temperature. If the star radiates like a black body the radiant flux M from unit area of its surface (emittance) will be given by:

$$M = \sigma T^4, \quad \ldots\ldots\ldots \quad (1)$$

where σ is Stefan's constant and T the temperature of the star. If R is the distance and r the radius of the star, then the flux density E received at the Earth will be:

$$E = \frac{4\pi r^2}{4\pi R^2} M = \frac{r^2}{R^2} \sigma T^4. \quad \ldots \quad (2)$$

Thus if we know T, R and E, we can find the radius r of the star.

This method requires that we know the distance of the star and this can only be measured for stars which are relatively close to us. As it happens there are very few *hot* stars for which the distance is known reliably and so their radii cannot be estimated reliably.

On the other hand, if the distance and the surface temperature are not known, we can find the surface temperature of a star by measuring its angular size Θ and the light flux density E; by substituting $\frac{1}{2}\Theta = r/R$ in eq. (2) we have:

$$E = \tfrac{1}{4}\Theta^2 \sigma T^4. \quad \ldots\ldots \quad (3)$$

The primary objective of the stellar interferometer at Narrabri is to measure the angular diameters of hot stars. The only existing measurements of angular diameter were made with Michelson interferometers at Mount Wilson. About 40 years ago an instrument was used with a maximum separation between the mirrors of about 6 metres. Its resolving power was only sufficient to measure six stars; all these stars are cool and relatively close. Ten years later a 15 metre instrument was used [2], but few new results seem to have been published, probably because of difficulties which will be explained later. These early observations have proved to be of considerable value and it is our aim at Narrabri to extend the work to the hot stars. The most valuable application of our data will depend on their accuracy and on what we find. At present we visualize their most important application to be the improvement of the scale of temperature for hot stars of types *O*, *B* and *A*. For reasons mentioned later the interferometer is limited to hot stars of spectral type *O*, *B*, *A* and *F* and

[1] The apparent magnitude m of a star is a measure, on a logarithmic scale, of the amount of light received from it. Thus, $m = -2.5$ log(light flux received) and the zero of the scale has been arbitrarily chosen as the mean magnitude of a group of Northern stars. On this scale the brightest star, Sirius, has a magnitude of about -1.5, while the faintest star which one can see with the naked eye has a magnitude of about $+6$. In the past stars were classified on a brightness scale and this scale has been adjusted so that a ratio of 100 to 1 in light flux corresponds to a difference in magnitude of 5.

[2] F. G. Pease, Ergebnisse der exakten Naturw. **10**, 84, 1931.

cannot be applied to cooler stars; furthermore, it is limited to stars brighter than apparent magnitude +2.5. It is expected that the instrument, in its present form, will be able to measure the angular diameters of about 50 hot stars.

In deriving equation (3) we have assumed that we can measure the total flux, integrated over all wavelengths, and that the star radiates like a black-body to which we can apply Stefan's law. For hot stars neither of these assumptions can be made. As we have already noted a substantial part of their light flux is absorbed by the Earth's atmosphere; furthermore, the spectral distribution of the radiation of hot stars does not follow closely that of a black body. In these cases the measurements of light flux and angular size are combined, using Planck's law, to find the equivalent black-body temperature of the star at the wavelength λ at which the light flux has been measured; this quantity is called the "*brightness temperature*" $T_b(\lambda)$ of the star. For a black body the brightness temperature and the effective temperature are obviously equal at all wavelengths. For the hot stars more sophisticated theoretical models are needed to relate them. The measured values of brightness temperature are then used to calibrate the theoretical models of hot stars to find their effective temperatures.

It is hoped in this way to find the effective temperatures of a number of hot stars of type *O*, *B* and *A* with an accuracy of better than 5% and thus to contribute significantly to our knowledge of the stellar temperature scale. In addition to these temperatures the observational programme for determining $\Theta = 2r/R$ should also yield the radii of the few hot stars for which reliable distances are known.

There are also some more specialized results which we hope to obtain. Firstly, we expect to discover that some of the well-known bright stars are unsuspected double or multiple stars. Many of the bright stars are already known to be multiple, either from direct observation of their components (visual binaries etc.) or from their spectra (spectroscopic binaries), but it is probable that the interferometer will reveal multiple stars which cannot be found by any other existing technique. Secondly, we propose to make measurements of the angular size of some stars which show bright emission lines and to check current theories of the origin of these lines. Finally, we shall try, if the equipment works sufficiently well, to measure the distribution of brightness over the disc of the bright star Sirius. This measurement, if it can be done, would give the first direct observation of the so-called "limb darkening" law for Sirius (the darkening towards the edge of the star disc), which is at present based almost entirely on theory.

Michelson's interferometer

Before describing the new instrument, let us consider the operation of the Michelson interferometer [3] and the reason why it cannot be extended to measure more stars.

Fig. 2 illustrates the principle of the interferometer. The light from the star received on the two small mirrors is directed into the focal plane of a large mirror where the two beams are brought together. The separation between the two small mirrors can be altered by the observer. When the light at the two small mirrors is mutually coherent the two beams of light interfere in

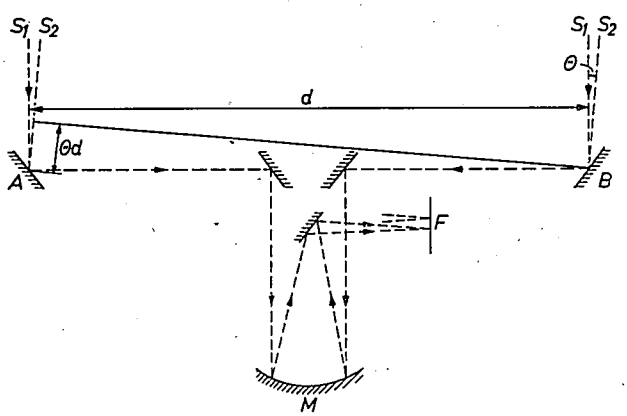


Fig. 2. The Michelson interferometer. Light from the star (or from two distant point sources S_1, S_2 separated by an angle Θ) is received at two small mirrors *A* and *B*. The two beams interfere in the focal plane *F* of the large mirror *M*. The modulation depth ("visibility") of the fringes in *F* is measured as a function of the mirror separation *d*.

the focal plane to form a pattern of alternate bright and dark bands, the "interference fringes". The "visibility" of these fringes, i.e. the depth of the modulation of the light intensity along the pattern relative to the mean light intensity, is a measure of the mutual coherence of the light at the two mirrors. When the star is not a point source the fringes belonging to different points of the source will overlap. This overlap, decreasing the visibility, depends both on the angular diameter Θ of the star and on the mirror separation *d*. When Θd is much smaller than the wavelength λ of the light the overlap is small and the visibility is large; but for $\Theta d \approx \lambda$, or greater, the visibility will decrease rapidly. The measurement of the angular diameter Θ of a star thus consists in observing the visibility as a function of the mirror separation *d*; from the value of *d* where the visibility is decreasing strongly: $d \approx \lambda/\Theta$, the angular diameter is obtained. An exact analysis [4] shows that the fringe visibility as a function of mirror separation is the Fourier transform of the angular intensity distribution across the stellar disc when reduced to an equivalent line source [5].

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The relation between visibility, mirror separation and angular diameter can readily be indicated more quantitatively for the simple case of two distant point sources S_1 and S_2 at an angular distance Θ, both emitting monochromatic light of the same wavelength λ (fig. 2). The light from S_1 only will produce fringes in the focal plane because the difference between the optical path lengths in the beams reaching a point in F by A and B respectively, depends upon the point in F considered. The centres of the bright bands of this pattern occur in points to which the optical paths differ by $n\lambda$; the dark band centres have path differences of $(n + \frac{1}{2})\lambda$. The light from S_2 will produce a second independent fringe pattern with the same band spacing. The difference of the two light paths from S_2 to a given point, however, differs by an amount Θd from the path difference for S_1. Thus the second fringe pattern is displaced by $\Theta d/\lambda$ of a fringe from the first and for $\Theta d/\lambda = 0, 1, 2, \ldots$ the visibility is a maximum, for $\Theta d/\lambda = \frac{1}{2}, \frac{3}{2}, \ldots$ it is a minimum (equal to zero when S_1 and S_2 are equally bright). Thus for two monochromatic point sources of the same wavelength the visibility is an oscillating function of Θd. When these considerations are extended to a monochromatic source of finite size results are obtained as those in *fig. 3* giving the visibility as a function of Θd for the case of an evenly radiating line source.

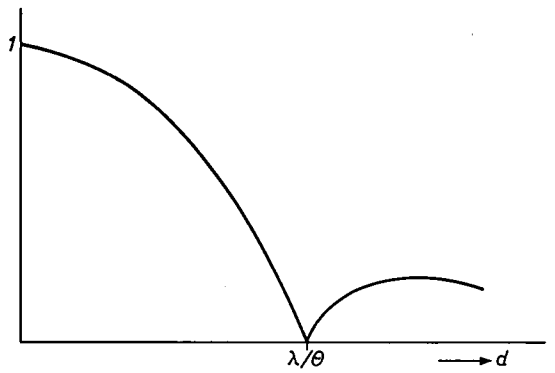


Fig. 3. The visibility of the fringe pattern in the Michelson interferometer is shown as a function of mirror separation d for an evenly radiating line source of angular size Θ emitting monochromatic light of wavelength λ. From the separation at which the visibility is zero the angular size can be derived.

The reason why the Michelson method *cannot* be used in practice with very large instruments to measure very small stars is connected with the fact that one cannot use an infinitely small frequency bandwidth. This introduces a finite critical length ΔL such that the number of wavelengths in ΔL differs by approximately one for the extreme frequencies of the band. If the path lengths from a point source to the focal plane along two beams differ by ΔL or more, these beams will no longer give rise to an interference pattern, because the fringes belonging to the individual frequencies overlap so strongly that no visibility remains. If the frequency bandwidth is $\Delta\nu$, then $\Delta L = n = (n-1)(\lambda + \Delta\lambda)$ from which we obtain $\Delta L = \lambda^2/\Delta\lambda$ or, with $\Delta\nu/\nu = \Delta\lambda/\lambda$, $\Delta L = c/\Delta\nu$ (c = velocity of light).

For a typical bandwidth $\Delta\lambda$ of 5 nm this critical length ΔL equals 0.06 mm and the two paths from the star to the focal plane should be equal to better than, say, 0.01 mm. This raises two difficulties in extending Michelson's interferometer to larger instruments. Firstly, attaining this accuracy from the small mirrors to the focal plane sets extremely close limits to any flexure or differential expansion of the two arms of the instrument; also to equalize the paths from the star to the mirrors to the same order of accuracy requires that the instrument should be "guided" very exactly at the star. As a result it is doubtful whether a Michelson interferometer could be built with a baseline of 100 metres, at least without inordinate difficulty and expense. Secondly, as with the telescope, a major difficulty is introduced by atmospheric scintillations. Movements of the air and the resultant pressure fluctuations vary, differentially, the optical lengths of the two beams from the star to the two small mirrors and change the paths of the beams, the latter effect giving an angular, or positional, scintillation to the star. Both effects move and distort the fringe pattern in the focal plane. This movement makes the fringes difficult to observe and introduces a subjective element into the results. It is to be expected that, by the use of modern photoelectric techniques, the effects of scintillation could be reduced and the observations made more objective; nevertheless, this has not yet been demonstrated.

The intensity interferometer

We now consider the intensity interferometer itself. A simplified outline of the intensity interferometer is shown in *fig. 4*. Light from the star is received on two separated mirrors each of which concentrates the light

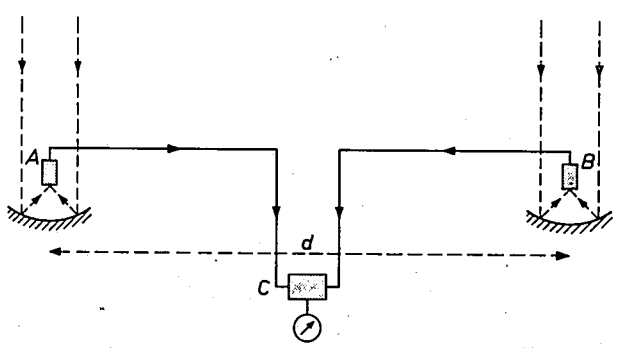


Fig. 4. Schematic diagram of the intensity interferometer. Light from the star is focused on two photoelectric detectors A and B. The correlation between the fluctuations in the output currents of A and B is measured in the correlator C, as a function of the detector separation d.

[3] C. Candler, Modern interferometers, Hilger and Watts, London 1951, p. 235.

[4] L. Mandel, Fluctuations of light beams, Progress in Optics, 181-248, 1963.

[5] "The equivalent line source" is a line, parallel to the line connecting the mirrors, for which each element has an intensity equal to the total intensity of the corresponding perpendicular strip of the disc.

on to a photoelectric detector at its focus. The output currents of these detectors contain high frequency fluctuations which are amplified and compared in an electronic correlator. The angular diameter of the star is found by measuring the correlation between these electrical fluctuations as a function of the separation between the two mirrors.

The intensity interferometer differs radically from Michelson's interferometer; the light beams received at the two mirrors are *not* brought together to produce direct optical interference but, in effect, the interference is carried out electrically. This novel technique has two valuable advantages. Firstly, it is not necessary to construct and to guide the instrument with the extreme mechanical precision which is required by a Michelson interferometer. Secondly, its operation is almost completely unaffected by atmospheric scintillation. It is therefore possible to construct an extremely large instrument which is capable of measuring the very small angles subtended by the hot stars.

A qualitative explanation

The principle of the intensity interferometer can be understood in a qualitative way as follows:

The light from a star arriving at the interferometer can be considered as the superposition of a large number of plane harmonic waves each characterized by a certain direction corresponding to one point in the light source, and a certain frequency and phase. The superposition of these waves results in light with a fluctuating intensity; that is to say, the electric vector of the light at the photocathodes has, as in the case of thermal noise, a fluctuating amplitude. Since the output currents from the detectors are proportional to the intensity of the light, the output currents will contain the same fluctuations.

In the case where the source of light is a *point source* the relative phases of all the harmonic components of the light will be the same at the two detectors, provided, of course, that the line joining the two detectors is normal to the direction of the light source. Under these conditions, it is obvious that the fluctuations in intensity at the two detectors, and therefore also in their output currents, must be completely correlated.

Let us now consider that the source of light is extended and that we can represent it by an array of point sources S_1, S_2, S_3, ... S_n, each point being separated from the next by the small angle $\Delta\Theta$; let Θ be the total angular diameter of the source (see *fig. 5*). If the line joining the two detectors is normal to the direction of S_1, then all the waves from S_1 will arrive at A and B with the same relative phases. The light from S_2, S_3, S_4 etc. however, will arrive at B *later* than at A by times $d\Delta\Theta/c$, $2d\Delta\Theta/c$ etc., so that the component waves from each of these sources have at B which are different from those at relative phases A. Thus, at any given instant they will combine to give different intensities, at the two detectors, and this will also hold for the combination of all contributions from S_1, S_2, S_3, ... S_n. The average intensities at A and B will, of course, be the same but the fluctuations of intensity will no longer be perfectly correlated.

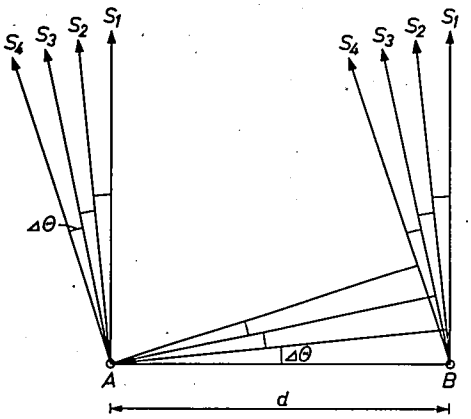


Fig. 5. The figure relates to the discussion of the fluctuations in the intensity of the light at two points A and B separated by a distance d, arriving from an array of point sources S_1, S_2, ..., separated by angles $\Delta\Theta$.

Provided that the maximum path difference Θd is small compared with the wavelength of the light, it is clear that there will be little difference between the light received at A and B. Thus, when A and B are close together ($d \ll \lambda/\Theta$) the correlation between the intensity fluctuations at the two detectors will be high; but as the baseline d increases and becomes comparable with λ/Θ the correlation will decrease and eventually will approach zero. In fact by measuring the correlation $\bar{C}(d)$ — which will be more closely defined below — as a function of d it is possible to find Θ, the angular diameter of the source of light.

The explanation given above is put in terms of light waves; however it applies to other ranges of wavelength as well, for example intensity interferometry has been used in radio astronomy.

The quantitative relationship between correlation, detector separation and angular diameter

A detailed calculation [6], which is too lengthy to be reproduced here, shows that the correlation $\bar{C}(d)$ expected when the two detectors are separated by a distance d is given by:

$$\bar{C}(d)/\bar{C}(0) = \Gamma^2(d). \quad \ldots\ldots \quad (4)$$

where $\bar{C}(0)$ is the correlation which would be observed with the detectors close together, and $\Gamma(d)$ is the modulus of the normalized Fourier transform of the angular distribution of intensity across the light source

when reduced to an equivalent line parallel to the baseline of the detectors. Thus we arrive at the strikingly simple result that the correlation observed with an intensity interferometer is proportional to the *square* of the fringe visibility in a Michelson interferometer with the same mirror separation. Thus the angular diameter of a star is determined by observing how the correlation decreases with increasing baseline and then finding the best fit with the theoretical curves of $\Gamma^2(d)$ for sources of different angular size and a given intensity distribution across the disc.

Fig. 6 shows the correlation as a function of d for two line sources corresponding to homogeneously radiating discs.

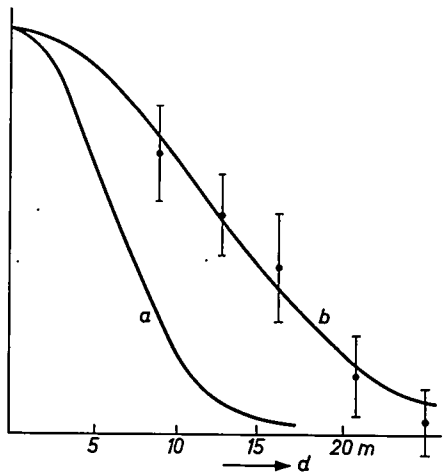


Fig. 6. The correlation as a function of detector separation d. The two curves are theoretical curves (eq. 4) for two sources corresponding to homogeneously radiating discs; *a*) disc angular size 0.008 sec; *b*) disc angular size 0.004 sec. The vertical line segments indicate typical experimental results and their probable error.

Correlation and noise

The correlation is the time-average of the cross-product of the detector current fluctuations. It is measured by applying the fluctuations, after these have been amplified and limited in frequency band, to a linear multiplier. The mean d.c. output of the multiplier is the correlation.

The correlation measurement is complicated by an unavoidable extra source of noise: the "*shot noise*" in the detector currents. Let us first consider the fluctuations in the output current of one detector. It has been shown [7] that, in the simple case where the incident light is linearly polarized and completely coherent over the surface of the detector, the mean square of these fluctuations measured at the input of the multiplying unit can be written:

$$\overline{N^2}=2e^2\left[\int_\nu A\alpha(\nu)n(\nu)\mathrm{d}\nu+\int_\nu A^2\alpha^2(\nu)n^2(\nu)d\nu\right]\int_f|F(f)|^2\mathrm{d}f, \quad . \; . \; . \quad (5)$$

where e is the electronic charge, A is the area of the detector, $n(\nu)$ is the number of quanta incident on unit area in unit time and unit bandwidth, $\alpha(\nu)$ is the quantum efficiency of the photocathode, and $F(f)$ is the frequency response of the complete electrical system up to the multiplier input and including the photoelectric detectors; the respective integrations are carried out over the optical (ν) and electrical (f) bandwidths. The first of the two terms in brackets represents the shot-noise in the cathode current of the photomultiplier. The second term represents the fluctuations in the current corresponding to the fluctuations of intensity in the incident light.

Let us now consider the outputs of two identical detectors. It is clear that the shot-noise in the two detectors will be uncorrelated, but it can be shown that the fluctuations due to the intensity fluctuations of the light, represented by the second term in (5), are correlated. In the simple case when the light is completely coherent over the two detectors together — when the separation $d = 0$ —, and again linearly polarized the correlation is given by [7]:

$$\bar{C}(0) = 2e^2\int_\nu A^2\alpha^2(\nu)n^2(\nu)\mathrm{d}\nu\int_f|F(f)|^2\,\mathrm{d}f. \quad . \; . \quad (6)$$

For zero separation the fluctuations in the light intensity, and thus the fluctuations in the output currents due to these, are identical for the two detectors. Therefore it is not surprising that the expression (6) is identical to the second part of (5): both expressions represent the time average of the product of two identical current fluctuations.

The signal to noise ratio

In equation (5) the second term represents the "signal", i.e. the fluctuations due to the light, which are correlated for two detectors close together; the first term represents the noise. If in the optical bandwidth the dependence of α and n upon ν can be neglected, then (5) shows that the ratio of signal power to noise power equals $A\alpha(\nu)n(\nu)$, i.e. the current signal to noise ratio is $\sqrt{A\alpha(\nu)n(\nu)}$. Aan is the number of photoelectrons per unit time per unit frequency bandwidth of the light. Now even for large mirrors and bright stars this number is very small (typically $\sim 10^{-4}$), and so in the output of the detectors the fluctuations due to the light are submerged in a much greater shot noise.

The output current of the multiplier is proportional to the product of the input currents. Therefore (cf. eq. 5 and 6), for the signal directly after the multiplication process, i.e. the correlation, the current signal to noise ratio now equals Aan. However, in practice, after the

[6] R. Hanbury Brown and R. Q. Twiss, Proc. Roy. Soc. A **243**, 291, 1957.
[7] R. Hanbury Brown and R. Q. Twiss, Proc. Roy. Soc. A **242**, 300, 1957.

multiplication process the noise is considerably reduced by the time-averaging process and the current signal to noise ratio is improved by a factor $\sqrt{T\Delta f}$ where T is the total time of observation and Δf is the electrical bandwidth of all the equipment between the photocathodes and the multiplying unit.

Also the influence of the light on the signal to noise ratio is contained in the factor Aan, which is the quantum efficiency of the photocathodes times the number of quanta received per unit time *per unit bandwidth of the light*. Thus the signal to noise ratio does not depend simply on the total light received but is also a function of the way in which the light is distributed over the optical bandwidth. In the simple case where this distribution is uniform, the signal to noise ratio is *independent* of the optical bandwidth provided only that it is much greater than the electrical bandwidth. A simplified derivation of these statements is given in the following article.

In practice, when we calculate the signal to noise ratio for an actual instrument, we find that a reasonable performance can only be achieved for even the brightest stars by using extremely large mirrors, high quantum efficiency photocathodes, wide electrical bandwidths and very long exposures. Furthermore we find that observations are restricted not only to *bright* stars but also to *hot* stars.

This last limitation is not immediately obvious and deserves a few words of explanation. In eq. (6) the correlation is given for the case where the size of the detectors, their separation and the angular diameter of the source are so small that the incident light can be considered as perfectly coherent over the surface of each detector and at the two detectors. In practice because the angular size of the source and the spacing between the detectors are not zero, the light will not be coherent over the two detectors. Furthermore the light may not be coherent over each reflector due to its size, i.e. each reflector may be large enough to partially resolve the source. Thus the correlation given by eq. (6) will be reduced by a factor which will depend on the size and spacing of the detectors and on the size and shape of the source of light. When this factor is included we find that the signal to noise ratio is a function not only of the parameters listed above and the brightness of the source, but also of its *brightness temperature*; in fact for a given time of observation there is a maximum possible signal to noise ratio, which increases as the brightness temperature increases, no matter how large the detectors are made. Thus a practical intensity interferometer is limited not only to stars *brighter* than a certain magnitude but also *hotter* than a certain minimum temperature; the exact values of these limits depending on the parameters of the instrument.

The way in which the brightness temperature enters into the signal to noise ratio, when the detectors are not small, is roughly as follows. The factor by which (6) must be reduced to obtain the correlation, if the light is not completely coherent over one detector, is determined by $\lambda/\Theta A^{1/2}$, analogous to $\lambda/\Theta d$ when the influence of the mirror separation d is considered ($A^{1/2}$ is roughly the mirror diameter). The reduction factor decreases as $\lambda/\Theta A^{1/2}$ decreases. For $\lambda/\Theta A^{1/2} \gg 1$ it equals unity (no reduction), for $\lambda/\Theta A^{1/2} \ll 1$ it can be shown [6][7] to approximate to $\lambda^2/\Theta^2 A$. In the latter case (large mirrors) the signal to noise ratio is proportional to $Aan\lambda^2/\Theta^2 A = a\lambda^2 n/\Theta^2$. Now $4n/\Theta^2 = nR^2/r^2$ (R distance, r radius of the star) equals the number of photons emitted per unit time per unit bandwidth *per unit area of the surface of the star* which is directly given by the brightness temperature. Thus in this case the brightness temperature and not the apparent magnitude determines the signal to noise ratio. In intermediate cases both quantities appear in the signal to noise ratio.

The tolerances in construction and alignment and the effect of atmospheric irregularities

The most significant advantages of the intensity interferometer are that it does not require precise mechanical construction or guiding and that it is substantially unaffected by atmospheric scintillation [8].

As we have seen it is essential to construct and guide Michelson's interferometer so that any difference between the light-paths through the two arms of the instrument does not exceed $c/\Delta\nu$, or, very roughly, one wavelength of the light. On the other hand in an intensity interferometer any path differences in the two arms must be small compared with the wavelength, not of the light, but of the electrical fluctuations transmitted from the photoelectric detectors to the correlator. More precisely the cross-correlation coefficient between the two sets of fluctuations produced at the two photocathodes is, as a function of relative time-delay, given by the Fourier transform of their power spectrum (Wiener-Khintchine theorem). Thus, assuming the two spectra to be identical, rectangular and extending to a maximum frequency f_{max}, the correlation will decrease considerably as soon as the difference in time delay between the two signals reaching the correlator exceeds a small fraction, say one tenth, of $1/f_{max}$ seconds. In principle, as we have already noted, it is desirable to use as large an electrical bandwidth as possible since the signal to noise ratio increases as the square root of the bandwidth; but in practice an upper limit of about 100 Mc/s is set by the response of the photoelectric detectors. Hence any differential time delays must not exceed about 10^{-9} s. This limitation applies to any relative time delays in the light reaching the two detectors as well as in the paths from the detector cathodes to the correlator. For the light this means, roughly speaking, that any path differences in the beams reaching the detectors must not exceed about 30 cm in air. In practice these tolerances are relatively

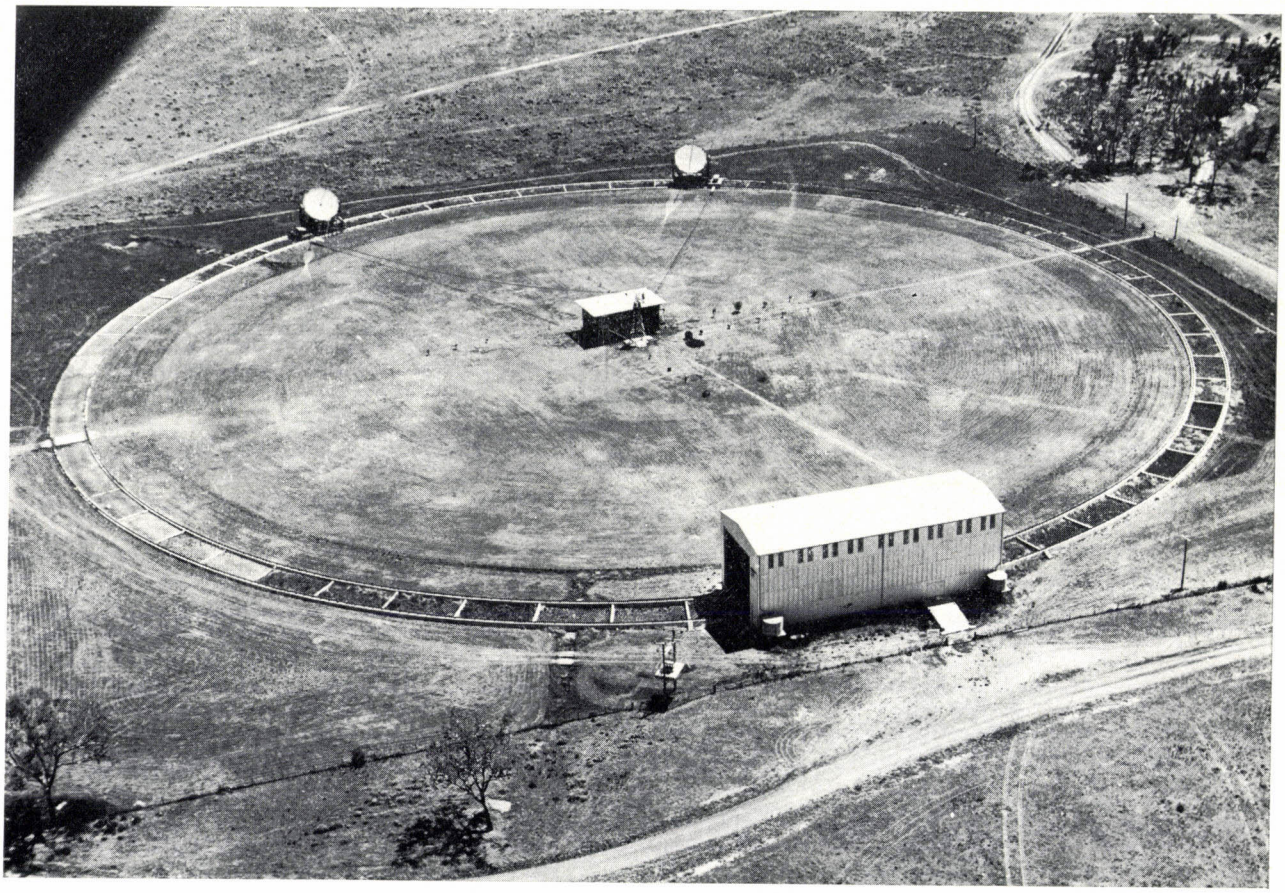


Fig. 7. The interferometer at Narrabri. The railway track has a diameter of 188 m. In the centre the control building and the cable tower, 9 m high. In the foreground is the parking garage for the reflectors.

easy to meet, even in a very large instrument with a baseline of a few hundred metres and it is therefore possible to construct an interferometer with a very high resolving power.

The second advantage of an intensity interferometer is its freedom from the adverse effects of atmospheric scintillation which are so serious in Michelson's interferometer. Briefly, the explanation of this remarkable property is contained in the previous paragraph. Thus, to affect the observed correlation significantly, the atmosphere must introduce differences in the relative time of arrival of the light at the two detectors of at least 10^{-9} s (equivalent to 30 cm in air). A detailed consideration of the types of atmospheric irregularity which are likely to exist and to be responsible for scintillation shows that the relative time delays to be expected are small compared with 10^{-9} s. Hence from a theoretical point of view it appears unlikely that atmospheric irregularities can modify the observed correlation, and this conclusion is confirmed by the limited amount of practical experience which we already have.

The interferometer at Narrabri

After a series of laboratory tests of the principle, a first attempt to make a stellar intensity interferometer was made in 1955. A pilot instrument was built, using Army searchlight mirrors with a diameter of 156 cm, and the angular size of Sirius was measured [9]. Following the experiment a larger instrument was built which is now installed and working at Narrabri Observatory. The Observatory is in flat pastoral country about 500 kilometres north of Sydney and has an elevation of about 200 metres above sea level. The site has reasonably clear weather, more than half of the nights are free from cloud, and the wind speed at night is usually less than 15 km/h. The layout of the installation is shown in *figs. 7* and *8*. The two large reflectors are mounted on trucks which run on a circular railway track with a diameter of 188 metres. The reflectors are controlled by a computer which calculates continuously the azimuth and elevation of the star. To follow the star in azimuth the reflectors move around the circle, and in

[8] R. Hanbury Brown and R. Q. Twiss, Proc. Roy. Soc. A **248**, 199, 1958.

[9] R. Hanbury Brown and R. Q. Twiss, Nature **178**, 1046, 1956. R. Hanbury Brown and R. Q. Twiss, Proc. Roy. Soc. A **248**, 222, 1958.

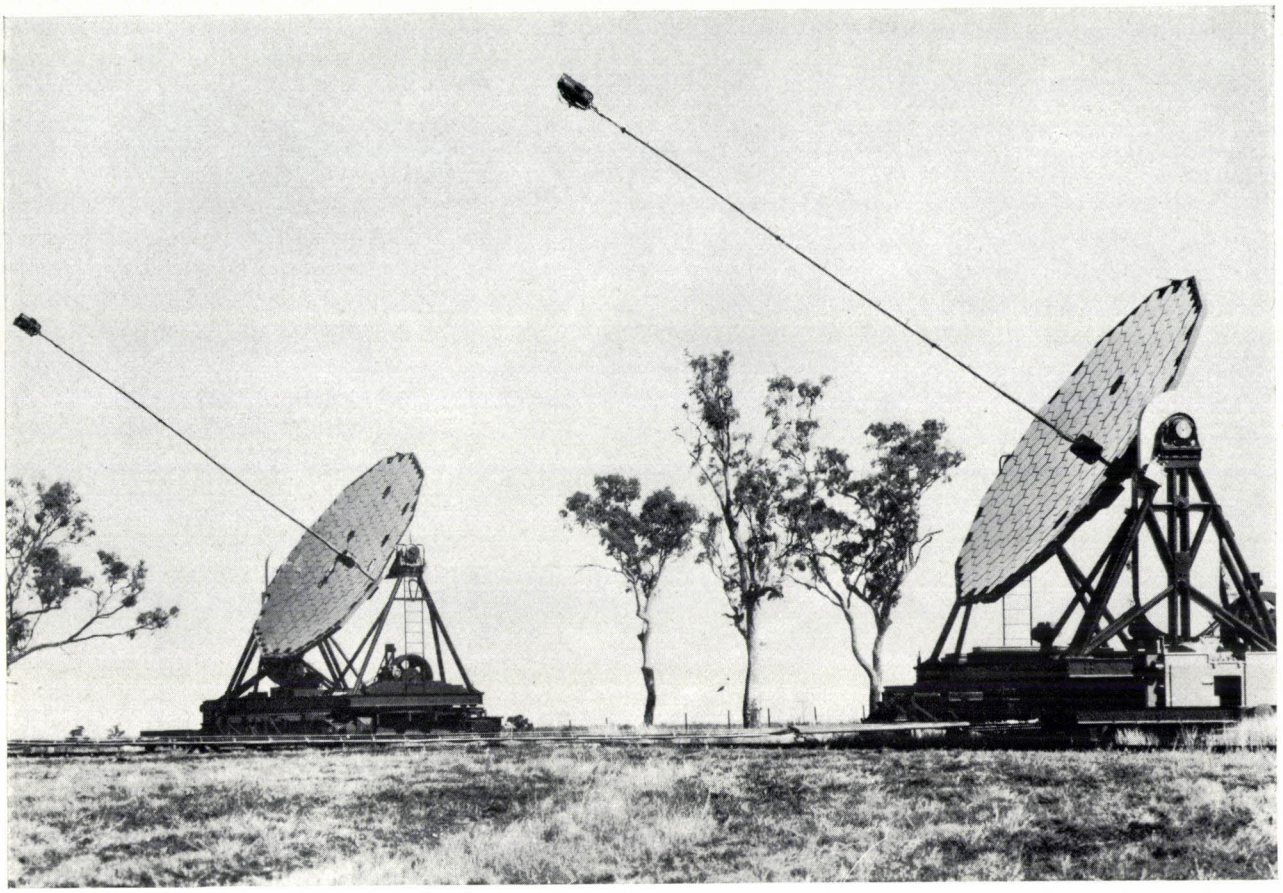


Fig. 8. The reflectors. Each reflector consists of 252 hexagonal mirrors and has a diameter of about 6.7 m. The pole carrying the photoelectric detector is 11 m long.

elevation they tilt about a horizontal axis. The separation between the reflectors can be varied from about 9 to 188 metres and they are arranged to move so that, at a constant separation, the line joining them is always at right angles to the direction of the star. This last point is an important feature of the design as it ensures that the difference in the times of arrival of the light at the two detectors is less than the limit explained above. In addition each reflector is mounted on a turntable on the truck and moves on this turntable so that it is always looking precisely in the direction of the star.

The reflectors are very large and crude by ordinary astronomical standards. Each has a light alloy framework about 6.7 metres in diameter on which 252 hexagonal mirrors are mounted. The framework is paraboloidal in shape to ensure that all of the light reaches the focus at approximately the same time, and each mirror is mounted on a three point suspension so that it can be set to reflect its beam on to the photoelectric detector. The mirrors are made of glass front-coated with aluminium and protected with silica. Each mirror has an electric heating pad cemented to it to prevent the formation of dew on the reflecting surface. Practical experience shows that 12 watts per mirror is sufficient during the winter at Narrabri. The mirrors have a spherical surface with a focal length appropriate to their position on the framework and averaging 11 metres.

At the focus the converging light is rendered parallel by a negative lens, passed through a narrow-band (8 nm) filter and focused through an iris diaphragm with a maximum aperture of 7.5 cm on to the photocathode of a 14 stage photomultiplier. The r.f. output from the photomultiplier is carried by a low loss coaxial cable to the correlator.

Although the computer is sufficiently accurate to point the reflectors at the star and keep the spot of light, which is 2.5 cm in diameter, on the photocathode, there is also a device for correcting their direction to keep the spot in the centre of the photocathode. This correction is necessary because the railway track is neither perfectly flat nor circular. The correction signal is provided by a second photomultiplier mounted alongside the main photomultiplier on each reflector. One of the 252 mirrors is set so that it reflects the light on to the centre of this guiding photomultiplier whilst the remainder illuminate the centre of the main photomultiplier. A D-shaped shutter rotates in front of the guiding photomultiplier (see *fig. 9*) and by interrupting the light

produces an a.c. output from the photomultiplier. Two lamps and photo-detectors derive, from the shutter, reference waves, in quadrature, which are used to rectify the photomultiplier output synchronously and to reveal the magnitude and direction of the azimuth and elevation errors. With this correcting system the r.m.s. guiding error with the reflectors in motion is about one minute of arc corresponding to about 6 mm deviation of the light spot at the focus.

In the centre of the track is a control building which houses the computer, the correlator, motor-generators and air-conditioning plant. The trucks are connected to the equipment in the control building by cables suspended from a steel catenary wire which is itself attached to a bearing at the top of a central mast. The radial pull of this wire, roughly two tons, is taken by a tender which is towed behind each mirror truck. In the southern part of the track there is a very large garage built over the railway track in such a way that the two mirrors can be protected completely from the weather when not in use. An interesting, but expensive, feature of the design is that the trucks can be parked inside the garage without detaching any of the cables; this is achieved by slots which run from the ends almost to the centre of the garage wall on the inner side of the track.

The computer is an electro-mechanical analogue computer which solves the equation connecting azimuth, elevation, hour angle and declination. There is not space in the present article to describe it in detail and it must suffice to say that it has an accuracy of about two or three minutes of arc, and has proved to be exceptionally reliable. The correlator is described in the following article.

The whole instrument was shipped from the U.K. to Australia towards the end of 1961 and made its first measurement of a star in July and August 1963. These first tests showed that, although all parts of the equipment were functioning properly, the sensitivity was approximately one stellar magnitude less than the designed value. The instrument was designed to give an r.m.s. correlation to noise ratio of 3/1 in one hour's observation of a star with a photographic magnitude of +2.5. In its present form it will only give this performance on a star of magnitude +1.5. The reasons for this discrepancy have been investigated and it is almost certain that, by the use of improved photomultipliers, the design performance will be achieved. In the meantime the instrument is fully occupied in measuring the angular size of the brightest stars. Roughly speaking it takes 15 to 20 hours of exposure with a very clear sky to obtain a complete set of results giving the size of one star. Among the stars that have been measured are Vega [10] (type *A*0), β Crucis (type *B*1) and Altair (type *A*7). The results on β Crucis are of considerable interest by themselves; they represent the first direct measurement of the angular size of a *B* star and they involved the use of baselines up to 100 metres in length. The probable error in the measurement of the angular size of these stars is a little less than $\pm$ 5 per cent.

Finally it can be said that the first results of this novel and interesting instrument are encouraging, and we look forward to carrying out a programme of observations which we hope will make a worthwhile contribution to our knowledge of the sizes and temperatures of the hot stars.

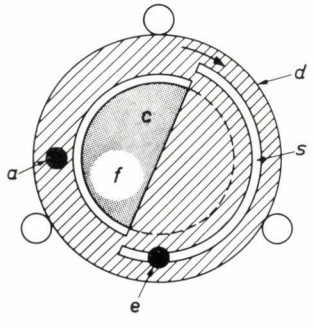


Fig. 9. Device for providing correction signals to correct the orientation of the reflectors. Light (*f*) from the star is focused by one of the 252 mirrors on to the photocathode *c* of a guiding photoelectric detector. This light is modulated by a rim driven shutter *d* if the light spot is not centred. Corrections for azimuth and elevation are distinguished by the phase of the modulation. The correction signals are obtained by synchronous rectification. The reference waves for this are obtained by interrupting two light beams (one for azimuth *a* and one for elevation *e*) passing through a slot *s* in the shutter *d*.

Acknowledgements

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[10] R. Hanbury Brown, C. Hazard, J. Davis and L. R. Allen, Nature **201**, 1111, 1964.

Summary. The stellar interferometer at Narrabri, Australia, has been set up to determine the angular diameters of some 50 bright stars. A knowledge of the angular diameter can be instrumental in determining the diameter and surface temperature of a star, and thus can greatly contribute to a better general understanding of the structure of stars. Up to now determinations of angular diameters have only been successful for a few stars, using Michelson's interferometer.

The intensity interferometer, which has been developed in the last 10 years, is rather similar to Michelson's interferometer. In the latter the light from the star is received at two separate plane

mirrors and the two beams are made to interfere: the visibility of the fringes decreases as the spacing of the mirrors is increased and the angular diameter can be determined from the rate of decrease of the fringe visibility. In the intensity interferometer the light from a star is received at two separate parabolic reflectors. The light is photoelectrically detected at the focus of each reflector, and the correlation between the fluctuations of the detector currents is measured. From the variations of correlation with mirror spacing it is possible to derive the angular diameter of the star. The advantage of the intensity interferometer over the Michelson interferometer is that the mechanical tolerances are very much greater; this means that a larger apparatus can be built and smaller stars can therefore be measured. Moreover, the sensitivity to atmospheric scintillations is much smaller.

The chief factor which limits the signal-to-noise ratio is shot noise in the photoelectric electrons. In each detector the fluctuations in the light intensity are very much smaller than the shot noise, which however is eliminated by the correlator. To obtain results the correlation has to be integrated over an appreciable period (minutes or hours), detectors of high sensitivity and fast response have to be used with large reflectors, and the measurements have to be limited to hot bright stars.

The reflectors of the Narrabri interferometer have a diameter of 6.7 metres, and their maximum spacing is 188 metres. Vega, β Crucis and Altair are among the stars whose angular diameters have been measured.

II. Measurement of the correlation

A. Browne

General considerations

In the previous article it has been shown that the angular diameter of a star may be obtained from the correlation between the fluctuations in the intensities of the light received at two detectors, as a function of the separation of these detectors. This article contains a description of the correlator (*fig. 10*), i.e. the electronic apparatus which measures this correlation.

A schematic outline of the apparatus is shown in *fig. 11*. The fluctuations in the output currents of the photodetectors, after being amplified and limited in frequency band, are multiplied together. By integrating the output current of the multiplying unit one obtains the average value representing the correlation. The input currents of the multiplying unit, X and Y, may be written as $X = x + z$, $Y = y + z$, where x, y and z are three independent noise currents. The components x and y are due to shot noise, and z is due to the common component in the light intensity fluctuations. The averaged product current is, because x, y and z are independent:

$$\overline{XY} = \overline{xy} + \overline{xz} + \overline{yz} + \overline{z^2} = \overline{z^2} = C\,.$$

This is the correlation, which must be measured for different separations of the detectors.

Signal to noise ratio

For an *infinite* integration time the averaging process separates the d.c. of the signal, $\overline{z^2}$, at the output of the multiplying unit, from the noise signals xy etc. because $\overline{xy}$ etc. $= 0$. For any *finite* integration time the noise is not eliminated completely and will cause an uncertainty in the measured correlation. The following simplified consideration shows how the signal to noise ratio depends on the integration time T and also on the bandwidth Δf and the mean light intensity. The component z is small with respect to x and y and the noise currents xz and yz are negligible when compared to xy. Referring to eq. (5) in the previous article, let us assume that $A\alpha n$ is independent of ν inside the optical frequency band from ν to $\nu+\Delta\nu$ and zero outside. In the same way, let F be independent of f inside the r.f. band from f to $f+\Delta f$ and zero outside. If we put $2e^2F^2\Delta\nu = \mathrm{k}$ and $A\alpha n = n_1$, then the two terms in (5) reduce to:

$$\overline{x^2} = \overline{y^2} = kn_1\Delta f, \quad \overline{z_0^2} = kn_1^2\Delta f. \quad . \; . \quad (7)$$

The latter term represents the mean square fluctuations due to the light in one detector (second term of eq. 5) as well as the correlation for zero separation (eq. 6). The integrating (averaging) process can be considered as the selection of signals from the output of the multiplying unit with frequencies in a bandwidth $\delta f \approx 1/T$ at $f = 0$. The noise xy is distributed mainly in the frequency bands 0 to Δf and $2f$ to $2f + 2\Delta f$. The latter band is of no importance. If we now assume that the noise power, which, apart from a resistance factor, equals $\overline{(xy)^2} = \overline{x^2}\,\overline{y^2} = \overline{x^2}\,\overline{y^2} = k^2n^2{}_1\Delta f^2$, is distributed evenly in the frequency band 0 to Δf, then the noise power in the band 0 to δf equals $k^2n_1^2\Delta f\,\delta f = k^2n_1^2\Delta f/T$. The correlation signal power is $(\overline{z_0^2})^2 = k^2n_1^4\Delta f^2$. Thus, for zero separation, we have:

$$\frac{\text{r.m.s. signal}}{\text{r.m.s. noise}} = \sqrt{\frac{\text{signal power}}{\text{noise power}}} = n_1\sqrt{T\Delta f}\,. \; . \quad (8)$$

Therefore, in a given integration time T, the signal to noise ratio increases as the electrical bandwidth Δf and the quantity n_1 (proportional to the light intensity per unit optical bandwidth, the effective cathode area and

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